

Take-Home Assignment weeks 37–38

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Exercise P5-1

The pipe is hoisted using the tongs. If the coefficient of static friction at A and B is μ_s , determine the smallest dimension b so that any pipe of inner diameter d can be lifted.

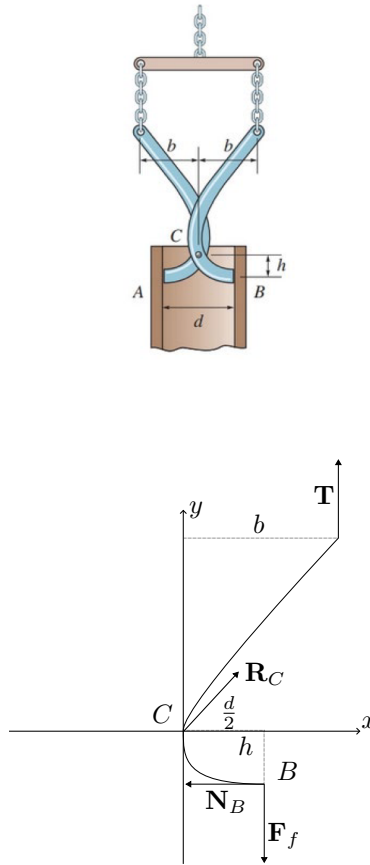


Figure 0.1: Free-body diagram of the right tong half

We consider one half of the tong, in this case the right half as shown on Figure 0.1. For this to be in equilibrium both the vector sum of all forces and moments about any point must be zero. I.e.

$$\mathbf{F}_R = \sum \mathbf{F} = \mathbf{0}$$

$$\mathbf{M}_O = \sum \mathbf{M}_O = \mathbf{0}.$$

For a couple moment not to arise, the lines of actions of all the forces must intersect at some common point, P , i.e. the forces must be concurrent. If the forces are not concurrent, there will always be some “perpendicular distance” between the lines of action of the forces and therefore a couple moment might be produced even if force equilibrium is required.

At point B , right at the verge of slippage, the line of action of the resultant force at B (which is the sum of the friction force and normal force components at this point) makes the friction angle, $\phi = \tan^{-1} \mu_s$ below the horizontal. We can therefore find the equation of the line using the formula for a line through a point, which in this case is $(-h, \frac{d}{2})$ and a slope, which in this case is μ as:

$$y + h = \mu_s \left(x - \frac{d}{2} \right) \implies y = \mu_s \left(x - \frac{d}{2} \right) - h.$$

The line of action at point B will lie along this line.

At the top the tong is attached to a vertical chain, such that the force that is transmitted to the tong at the top is vertical, with a line of action of $x = b$.

As mentioned above, all the lines of actions of the forces must intersect at a common point P . This means that the two lines defined above must intersect at this point P . The intersection between the line of action of $\mathbf{T}$ and the line of action of $\mathbf{R}_B$ can now be found as:

$$P = \left(b, \mu_s \left(b - \frac{d}{2} \right) - h \right).$$

The pin force $\mathbf{R}_C$ at point C will act along the line CP . If this point P is below point C one can imagine that the tongues will “open up” and not press against the wall. Therefore, for the jaws to wedge and hold, the concurrency point P must be at or above C as:

$$\begin{aligned} y_p &\geq 0 \\ \mu_s \left(b - \frac{d}{2} \right) - h &\geq 0 \\ \implies \mu_s \left(b_{\min} - \frac{d}{2} \right) - h &= 0 \\ b_{\min} &= \frac{h}{\mu_s} + \frac{d}{2}. \end{aligned}$$

Therefore the smallest dimension of b such that the pipe can be lifted is $b_{\min} = \frac{h}{\mu_s} + \frac{d}{2}$.

Exercise P6-1

The cart supports the uniform crate having a mass of 85 kg. Determine the vertical reactions on the three casters at A , B , and C . The caster at B is not shown. Neglect the mass of the cart.

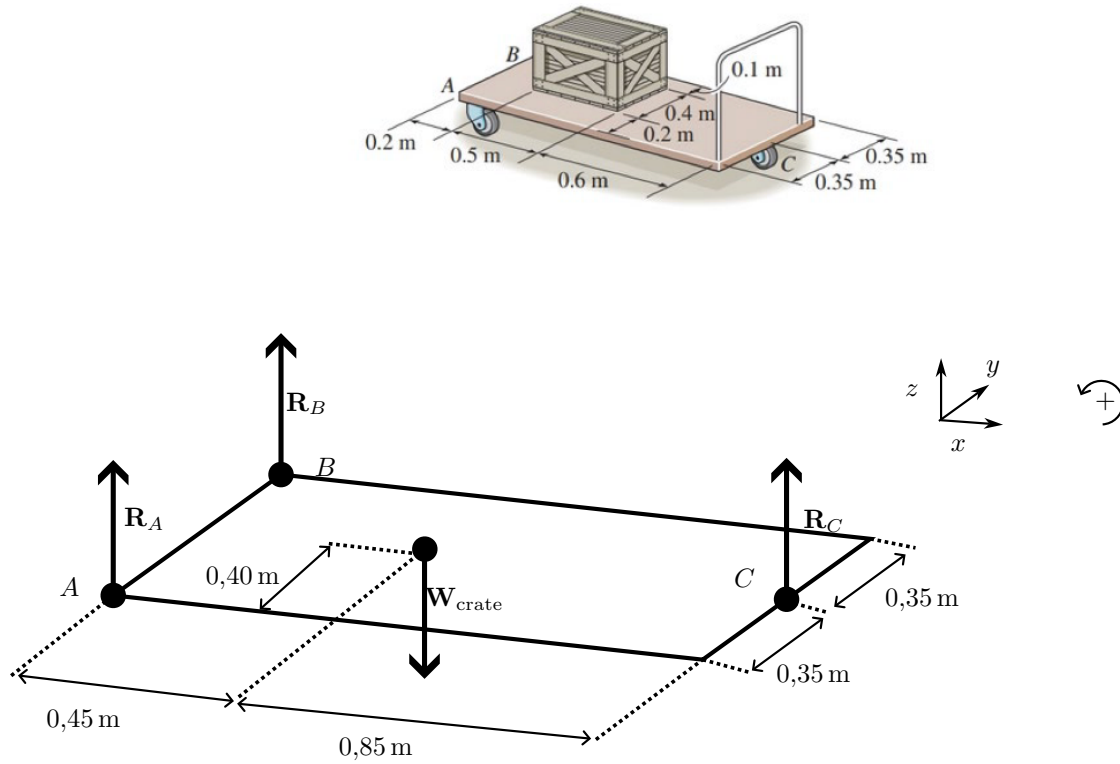


Figure 0.2: Free body diagram

A free body diagram of the cart as shown on Figure 0.2. For the cart to be in equilibrium the equilibrium equations must be satisfied. In three dimensions these may be:

$$\begin{aligned}\sum F &= R_A + R_B + R_C + W_{\text{crate}} = 0 \\ \sum M_{A_x} &= |AB|_x R_B + |AW|_x W_{\text{crate}} + |AC|_x R_C \\ \sum M_{A_y} &= |AB|_y R_B + |AW|_y W_{\text{crate}} + |AC|_y R_C.\end{aligned}$$

First we find the size of W_{crate} using Newton's second law as

$$W_{\text{crate}} = 85 \text{ kg} \cdot (-9,81 \frac{\text{m}}{\text{s}^2}) = -833,85 \text{ N}.$$

This weight will be acting from the center of the crate as it is assumed to be homogeneous. The distances

can be read of the free body diagram to be

$$\begin{aligned}
 |AB|_x &= 0 \\
 |AW|_x &= 0,45 \text{ m} \\
 |AC|_x &= 1,30 \text{ m} \\
 |AB|_y &= 0,70 \text{ m} \\
 |AW|_y &= 0,40 \text{ m} \\
 |AC|_y &= 0,35 \text{ m}.
 \end{aligned}$$

Hence

$$\begin{aligned}
 R_A + R_B + R_C &= 833,85 \text{ N} \\
 1,30 \text{ m} \cdot R_C &= 0,45 \text{ m} \cdot 833,85 \text{ N} \\
 0,70 \text{ m} \cdot R_B + 0,35 \text{ m} \cdot R_C &= 0,40 \text{ m} \cdot 833,85 \text{ N}.
 \end{aligned}$$

Solving for R_C can easily be done as

$$R_C = \frac{0,45 \text{ m} \cdot 833,85 \text{ N}}{1,30 \text{ m}} = 288,64 \text{ N}.$$

Now, solving for R_B becomes trivial as

$$R_B = \frac{0,40 \text{ m} \cdot 833,85 \text{ N} - 0,35 \text{ m} \cdot 288,64 \text{ N}}{0,70 \text{ m}} = 332,17 \text{ N}.$$

Lastly, R_A can be found to be

$$R_A = 833,85 \text{ N} - 288,64 \text{ N} - 332,17 \text{ N} = 213,04 \text{ N}.$$

Hence the three reaction forces will have sizes $R_A = 213 \text{ N}$, $R_B = 332 \text{ N}$, and $R_C = 289 \text{ N}$ and they will all be acting vertically upwards in the $+z$ -direction. This result makes sense as so far as the reaction force being highest for the caster closest to the crate (B), a little smaller for the sole caster a little further from the crate (C) and smallest for the caster at A , which is both somewhat far from the crate and more supported by the caster at B than was the case for the caster at C .

Exercise P6-2

Both pulleys are fixed to the shaft and as the shaft turns with constant angular velocity the power of pulley A is transmitted to pulley B . Determine the horizontal tension $\mathbf{T}$ in the belt on pulley B and the x , y , z components of reaction of the journal bearing C and thrust bearing D if $\theta = 45^\circ$. The bearings are in proper alignment and exert only force reactions on the shaft.

